INDUSTRIAL CUM SOFTWARE TRAINING REPORT

SUBMITTED IN PARTIAL FULFILLMENT OF THE DEGREE

OF

BACHELOR OF TECHNOLOGY

IN

COMPUTER SCIENCE AND ENGINEERING

SUBMITTED BY

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Under the Guidance of:

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Designation: Assistant Professor

Department: Computer Science & Engineering

Organization: Gulzar Group of Institutes

GULZAR GROUP OF INSTITUTES, LUDHIANA, PUNJAB

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TRAINING REPORT

UNDERTAKEN AT

FUTURE FINDER PVT. LTD

JANUARY 2026 TO JUNE 2026

ON

PLANT LEAF DISEASE DETECTION SYSTEM

SUBMITTED IN PARTIAL FULFILLMENT OF THE DEGREE

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DECLARATION

I hereby declare that the project work entitled ("Plant Leaf Disease Detection System") is an authentic record of my own work carried out at (Place of work) as requirements of four months Industrial Training for the award of degree of B.Tech Computer Science and Engineering, Gulzar Group of Institutes, Ludhiana, under the guidance of MR.Rohan Kashyap and Er.Manpreet kaur, during January 2026 to June, 2026.

I hereby declare that the above statement made by me is correct to the best of our knowledge and belief

(Signature of student)
Ravi Kumar
(2339282)

Date: ___________________

Abstract

The Plant Leaf Disease Detection System is a deep learning-based application developed to automatically identify and classify plant leaf diseases using Convolutional Neural Networks (CNN). The project was built using Python, TensorFlow, Keras, and Streamlit to provide an efficient solution for early crop disease detection. The model was trained on the PlantVillage dataset containing over 54,000 RGB images across 38 disease categories and 14 plant species. Various preprocessing techniques such as data augmentation, normalization, resizing, and max pooling were applied to improve model accuracy and generalization. The cnn model effectively detects visual symptoms including leaf spots, discoloration, lesions, and texture variations. A user-friendly Streamlit web application was developed to allow users to upload leaf images and receive real-time disease predictions instantly. The system helps reduce dependency on manual disease inspection and supports early preventive action in agriculture. This project demonstrates the practical application of deep learning and artificial intelligence in precision farming and smart agriculture.

Acknowledgement

I would like to express my sincere gratitude to all those who supported and guided me throughout the development of the Plant Leaf Disease Detection System using cnn. I am thankful to my teachers and mentors for their valuable guidance, encouragement, and technical support during the project. Their suggestions and feedback helped me improve the quality of this work.

I would also like to thank the developers and contributors of Python, TensorFlow, Keras, and Streamlit for providing powerful tools and frameworks that made this project possible. Special thanks to the creators of the PlantVillage dataset for providing a comprehensive dataset for training and testing the model.

Finally, I am grateful to my family and friends for their continuous motivation and support throughout the project development process.

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of the above training, we have found him/her a hardworking

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Chapter 1

Profile of Institute

1.1 Introduction to Future Finder

Future Finder is a professionally managed IT training and career development institute located in Mohali, Chandigarh. Established with a clear mission to bridge the gap between academic education and industry-ready skills, Future Finder has emerged as a trusted name in the technology training landscape of the Tricity region (Chandigarh, Mohali, and Panchkula). The institute specialises in providing practical, project-based training in domains such as Artificial Intelligence, Machine Learning, Python programming, Web Development, and Data Science. It caters to engineering students, graduates, and working professionals seeking to upgrade their technical competencies and enhance their employability in the competitive IT sector.

Future Finder has successfully mentored hundreds of students from leading engineering institutions across Punjab and Haryana, including partnerships with universities affiliated to Punjab Technical University (PTU) and Chandigarh University. The institute follows a hands-on, industry-aligned training methodology where each student is assigned a real-world project from the outset, ensuring that theoretical concepts are immediately reinforced through practical implementation.

1.2 Vision and Mission

1.2.1 Vision

To be the most impactful technology training institute in North India, producing industry-ready professionals who can confidently contribute to the rapidly evolving global IT ecosystem.

1.2.2 Mission

To empower students with hands-on technical skills, mentorship from industry practitioners, and exposure to real-world projects, enabling them to secure meaningful careers in technology and contribute positively to society through innovation.

1.3 Institute at a Glance

Table 1:1 Future Finder – Institute at a Glance

Parameter	Details
Name of Institute	Future Finder
Location	Mohali, Chandigarh, Punjab
Type	IT Training & Career Development Institute
Training Domains	AI/ML, Python, Web Development, Data Science, Cloud
Target Audience	Engineering Students, Graduates, Working Professionals
Training Mode	Offline (Lab-Based) and Online
Affiliation / Tie-ups	Punjab Technical University (PTU) affiliated colleges
Placement Assistance	Yes – Resume Building, Mock Interviews, Job Referrals

1.4 Location and Infrastructure

Future Finder is centrally located in Mohali (SAS Nagar), adjacent to Chandigarh – one of India's most well-planned cities and a growing IT hub. The institute is easily accessible from all major sectors of Chandigarh and Mohali via public and private transport, making it convenient for students from across the Tricot region.

The institute is equipped with modern computing laboratories featuring high-performance workstations installed with the latest software tools and development environments. Infrastructure highlights include:

- Dedicated computer labs with uninterrupted high-speed internet connectivity.
- GPU-enabled systems for deep learning and computer vision training sessions.
- A fully stocked digital resource library with access to e-books, research papers, and online course platforms.
- Seminar and presentation rooms for project reviews, industry talks, and technical workshops.
- Individual workstations assigned to each intern for the duration of their training period.

1.5 Courses and Training Programs Offered

Future Finder offers a diverse portfolio of short-term and semester-long training programs designed to meet current industry demands. The curriculum is regularly updated in consultation with industry mentors to ensure relevance and quality.

Table 1:2 Courses and Training Programs at Future Finder

S. No.	Course / Program	Duration	Mode
1	Python Programming (Basic to Advanced)	4–6 Weeks	Offline
2	Artificial Intelligence & Machine Learning	2–4 Months	Offline
4	Web Development (HTML, CSS, JS, React)	2 Months	Offline
5	Data Science & Data Analytics	3–6Months	Offline
6	Database Management (MySQL)	1 Month	Offline
7	Cloud Computing (AWS Fundamentals)	1 Month	Online

1.6 Achievements and Recognition

Since its inception, Future Finder has built a strong reputation for delivering quality training and tangible career outcomes. Key achievements include:

- Successful placement assistance provided to over 85% of trained students in reputed IT organisations across Chandigarh, Mohali, and beyond.
- Consistently high student satisfaction ratings, with trainees reporting significant improvement in technical confidence and project-building capabilities.
- Recognised as a preferred training partner by multiple B.Tech institutions in Punjab for their mandatory six-month internship programmes.
- Alumni placed in companies including Infosys, Wipro, TCS, HCL, and various Chandigarh-based IT startups working in domains of AI, web development, and cloud services.

1.7 Training Methodology

Future Finder follows a structured, project-first training methodology that distinguishes it from conventional coaching institutes. The approach is built on four pillars:

1.7.1 Concept Building

Each training module begins with a thorough conceptual introduction covering theory, algorithms, and real-world context. Instructors use case studies and visual aids to ensure students develop a strong foundational understanding before moving to implementation.

1.7.2 Hands-On Laboratory Sessions

Every concept is immediately followed by guided lab sessions where students implement the concepts on real datasets and development environments. This ensures active learning and immediate application of knowledge.

1.7.3 Live Project Assignment

Students are assigned industry-relevant projects from the beginning of their training. For the current batch, projects include domains such as plant disease detection using CNN, sentiment analysis, object detection systems, and e-commerce recommendation engines.

Chapter 2

Introduction

2.1 Introduction of Leaf Disease Detection

Agriculture is one of the most important sectors for the growth of the economy and food production. Plant diseases greatly affect crop quality and productivity, causing major losses to farmers and the agricultural industry. Traditional methods of detecting plant diseases mainly depend on manual inspection by experts, which is time-consuming, expensive, and sometimes inaccurate. With the advancement of Artificial Intelligence (AI) and Deep Learning technologies, automated disease detection systems have become an efficient solution for improving agricultural productivity and crop monitoring. The *Plant Leaf Disease Detection System using CNN* is a deep learning-based application developed to automatically identify and classify diseases in plant leaves from images. The project uses Convolutional Neural Networks (CNN), which are highly effective for image classification and pattern recognition tasks. The model was trained using the publicly available PlantVillage *dataset*, containing more than *54,000 RGB* images across *38* disease categories and *14 plant species*. The system was developed using *Python*, TensorFlow, *Keras*, and Streamlit to provide a user-friendly interface for real-time disease prediction. Users can upload images of plant leaves, and the system predicts the disease category along with the plant type. This project demonstrates the practical use of deep learning in agriculture and helps farmers take preventive measures at an early stage to reduce crop losses and improve productivity.

To improve the overall performance and accuracy of the model, various image preprocessing and enhancement techniques were applied during training. These techniques include image resizing, normalization, data augmentation, and max pooling operations, which help the model generalize better and accurately classify leaf images under different environmental conditions. The CNN architecture was specifically designed to learn and extract meaningful visual features from plant leaf images, enabling efficient disease recognition with high predictive accuracy.

This project demonstrates the practical application of deep learning and artificial intelligence in the field of agriculture. It highlights how AI-powered systems can support precision farming, smart agriculture, and automated crop monitoring. The proposed system aims to contribute toward sustainable agricultural development by improving productivity, reducing crop losses, and promoting the adoption of intelligent farming technologies for better crop management and disease control.

2.1.1 Importance of Agriculture

Agriculture is one of the most important sectors contributing to the economic growth and development of a country. It plays a vital role in providing food, employment, and raw materials for industries. A large percentage of the world's population depends directly or indirectly on agriculture for their livelihood. In developing countries, agriculture forms the backbone of the economy and significantly contributes to national income. Healthy crop production is essential to meet the increasing food demands of the growing global population. However, crop productivity is highly affected by several environmental and biological factors, among which plant diseases are one of the major challenges faced by farmers.

Plant diseases reduce both the quality and quantity of agricultural products. Diseases caused by bacteria, fungi, viruses, and pests spread rapidly and damage crops at different stages of growth. These infections not only reduce crop yield but also lead to major economic losses for farmers and the agricultural industry. Therefore, early detection and proper management of plant diseases are essential to improve agricultural productivity and ensure food security.

2.1.2 Challenges in Traditional Plant Disease Detection

Traditionally, plant diseases are detected through manual observation and visual inspection by agricultural experts or farmers. This process requires significant knowledge, experience, and continuous field monitoring. Although traditional methods are useful in some cases, they involve several limitations. Manual inspection is time-consuming, labor-intensive, and expensive, especially for large agricultural fields. In many rural and remote areas, expert guidance may not be easily available, making disease diagnosis difficult for farmers.

Another challenge associated with traditional disease detection is the possibility of human error. Different diseases often show similar symptoms such as spots, discoloration, and wilting, making accurate diagnosis difficult. Delayed identification of diseases can lead to rapid spread of infections, resulting in severe crop damage and productivity loss. These challenges highlight the need for intelligent and automated systems capable of detecting plant diseases accurately and efficiently.

2.1.3 Role of Artificial Intelligence in Agriculture

The rapid advancement of Artificial Intelligence (AI), Machine Learning (ML), and Deep Learning technologies has transformed several industries, including agriculture. AI-based systems are increasingly being used to improve farming practices, monitor crop health, automate irrigation, and optimize agricultural productivity. In the field of plant disease detection, AI techniques provide automated and accurate solutions for identifying diseases from leaf images.

Deep Learning, a subset of Artificial Intelligence, has gained significant popularity due to its ability to process large amounts of image data and automatically extract important features. Convolutional Neural Networks (CNNs), one of the most powerful deep learning algorithms, are widely used for image classification and object recognition tasks. CNN models can analyze leaf images and identify disease symptoms such as spots, lesions, texture changes, and color variations with high accuracy. These capabilities make CNN-based systems highly effective for automated plant disease detection.

2.1.4 Overview of the Proposed System

The Plant Leaf Disease Detection System using CNN is a deep learning-based application designed to automatically identify and classify diseases in plant leaves using image analysis techniques. The proposed system utilizes Convolutional Neural Networks (CNNs) to classify healthy and diseased leaves across multiple crop categories. The model was trained using the publicly available PlantVillage dataset, which contains more than 54,000 RGB images belonging to 38 disease categories and 14 plant species. The dataset includes healthy and diseased leaf samples from crops such as tomato, potato, apple, grape, pepper, strawberry, and others. The system was developed using Python, TensorFlow, Keras, and Streamlit. Python was used as the programming language for implementing the deep learning model, while TensorFlow and Keras were used for building and training the CNN architecture. Streamlit was used to develop a user-friendly web interface for real-time disease prediction. Users can upload plant leaf images through the web application, and the trained model predicts the disease category along with the corresponding crop type.

2.1.5 Image Preprocessing Techniques

Image preprocessing plays a crucial role in improving the performance and accuracy of deep learning models. Raw images collected from datasets often contain variations in size, lighting conditions, orientation, and quality. To handle these variations, several preprocessing techniques were applied before training the CNN model.

The first preprocessing step involves image resizing, where all leaf images are converted into a fixed dimension suitable for model training. Normalization techniques are then applied to scale pixel values within a standard range, helping the model converge faster during training. Data augmentation techniques such as rotation, flipping, zooming, and shifting are used to artificially increase the dataset size and improve model generalization. These techniques help the model perform well even when tested on unseen data.

Max pooling operations were also applied within the CNN architecture to reduce computational complexity and extract important image features. By using these preprocessing and enhancement techniques, the proposed system achieves better disease classification performance and improved prediction accuracy.

2.1.6 Convolutional Neural Networks (CNN)

Convolutional Neural Networks (CNNs) are specialized deep learning architectures designed for image processing and classification tasks. CNNs automatically learn important features from images through multiple layers such as convolution layers, activation layers, pooling layers, and fully connected layers. Unlike traditional machine learning methods, CNNs do not require manual feature extraction, making them highly efficient for image analysis. In this project, the CNN model was designed to identify visual patterns associated with plant diseases. The convolution layers extract low-level and high-level features such as edges, textures, shapes, and lesions from leaf images. Activation functions such as ReLU introduce non-linearity into the model, enabling it to learn complex patterns effectively. Pooling layers reduce image dimensions while preserving important features, and fully connected layers perform the final classification of diseases.

The CNN model was trained on thousands of leaf images to learn disease-specific characteristics. During training, the model continuously adjusted its weights and parameters to minimize classification errors and improve prediction performance.

2.2 Applications

The Plant Leaf Disease Detection System has various applications in the field of agriculture and smart farming. The system helps farmers detect plant diseases at an early stage, allowing them to take timely preventive actions and reduce crop damage. It supports precision agriculture by enabling intelligent crop monitoring using AI-based technologies. The project can also be used in agricultural research to study disease patterns and improve crop management techniques. The Streamlit-based web application allows users to upload leaf images and receive real-time disease predictions, making the system suitable for remote diagnosis and digital farming solutions. Additionally, the project reduces the need for manual inspection, saves time and cost, and contributes to the development of smart agricultural systems integrated with AI and IoT technologies.

2.2.1 Early Disease Detection

One of the major applications of the Plant Leaf Disease Detection System is early disease identification. Early detection helps farmers recognize infections before they spread widely across crops. By identifying diseases at an early stage, farmers can apply appropriate treatments and preventive measures, reducing crop damage and improving productivity.

2.2.2 Precision Agriculture

The proposed system supports precision agriculture by enabling intelligent crop monitoring using AI-based technologies. Precision farming focuses on optimizing agricultural resources and improving farming efficiency through data-driven decision-making. Automated disease detection systems help monitor crop health accurately and reduce unnecessary pesticide usage.

2.2.3 Smart Farming Solutions

The project contributes to the development of smart farming technologies by integrating Artificial Intelligence with agriculture. The system can be combined with IoT devices, drones, and cloud-based platforms for large-scale crop monitoring and automated farming solutions.

2.2.4 Agricultural Research

Researchers and agricultural scientists can use the system to study disease patterns and analyze crop health data. AI-powered disease detection systems help improve agricultural research by providing accurate and automated classification results.

2.2.5 Remote Disease Diagnosis

The Streamlit-based web application enables remote disease diagnosis by allowing users to upload leaf images from any location. This feature is especially useful for farmers in rural areas where agricultural experts may not be readily available.

2.3 Objectives

The main objective of this project is to develop an automated and intelligent plant disease detection system using Convolutional Neural Networks (CNN). The project aims to accurately classify plant leaf diseases from uploaded images by applying deep learning techniques and image preprocessing methods. Another objective is to improve model performance and prediction accuracy through data augmentation, normalization, and max pooling techniques. The project also focuses on developing a user-friendly Streamlit web application for real-time disease prediction and easy accessibility. Furthermore, the system is designed to assist farmers and agricultural researchers in early disease diagnosis, reduce crop losses, improve agricultural productivity, and demonstrate the practical application of AI and deep learning in agriculture.

2.3.1 Main Objective

The primary objective of this project is to develop an automated and intelligent plant disease detection system using Convolutional Neural Networks (CNNs). The system aims to accurately identify and classify plant diseases from leaf images.

2.3.2 Accurate Disease Classification

Another objective of the project is to improve disease classification accuracy using deep learning algorithms and image preprocessing techniques. The model is trained to recognize multiple disease categories across different plant species.

2.3.3 Real-Time Prediction System

The project aims to develop a user-friendly web application capable of providing real-time disease predictions. The Streamlit interface allows users to upload leaf images and receive instant prediction results.

2.3.4 Reduction of Crop Losses

The system is designed to help farmers detect diseases early and take preventive actions to reduce crop losses. Early diagnosis improves crop quality and increases agricultural productivity.

2.3.5 Promotion of AI in Agriculture

Another objective is to demonstrate the practical application of Artificial Intelligence and Deep Learning in agriculture. The project highlights how AI technologies can improve farming practices and support sustainable agricultural development.

2.4 Motivation

The motivation behind this project comes from the growing challenges faced by farmers due to plant diseases and the lack of quick and accurate disease diagnosis methods in many rural areas. Manual disease detection requires expert knowledge and regular field inspection, which may not always be available or affordable for farmers. As a result, diseases often spread before proper treatment can be applied, leading to significant crop losses and economic damage. With the rapid advancement of Artificial Intelligence and Deep Learning, there is a strong opportunity to build automated systems capable of identifying plant diseases with high accuracy. This project was motivated by the need to create a smart, efficient, and scalable solution that can support farmers in monitoring crop health and taking preventive actions at an early stage.

2.4.1 Increasing Plant Diseases

One of the major motivations behind this project is the increasing number of plant diseases affecting crop production worldwide. Climate change, environmental pollution, and changing agricultural practices have increased the spread of crop infections.

2.4.2 Limitations of Manual Detection

Traditional disease detection methods are time-consuming and require expert knowledge. Many farmers cannot afford regular expert consultations, leading to delayed disease identification and increased crop losses.

2.4.3 Advancement of Deep Learning

The rapid development of Artificial Intelligence and Deep Learning technologies motivated the development of intelligent disease detection systems. CNN models have shown remarkable performance in image classification and pattern recognition tasks.

2.4.4 Need for Smart Agriculture

Modern agriculture requires smart and automated solutions for efficient crop monitoring and management. AI-based systems help farmers improve productivity while reducing costs and manual efforts.

2.4.5 Sustainable Agricultural Development

The project is motivated by the need to support sustainable agricultural practices through technology-driven solutions. Automated disease detection systems help reduce pesticide overuse, minimize crop losses, and improve food production efficiency.

Chapter 3:

Literature Survey

SL. NO	Author	Title	Aim	Method	Parameter	Data Size	Result
1	Ferentinos (2018)	Deep Learning Models for Plant Disease Detection and Diagnosis	To detect and classify plant diseases using deep learning models	Convolutional Neural Network (CNN)	Accuracy, Loss Function, Epochs	87,848 images from PlantVillage dataset	Achieved 99.53% classification accuracy for plant disease detection
2	Brahimi et al.	Deep Learning for Tomato Diseases: Classification and Symptoms Visualization	To classify tomato leaf diseases automatically	AlexNet and GoogLeNet CNN architectures	Accuracy, Feature Extraction, Image Classification	14,828 tomato leaf images	Achieved 99.18% accuracy in tomato disease classification
3	Mohanty et al. (2016)	Using Deep Learning for Image-Based Plant Disease Detection	To identify plant diseases from leaf images using deep learning	Deep CNN with Transfer Learning	Precision, Recall, Accuracy	54,306 images across 38 classes	Achieved 99.35% accuracy on controlled dataset images
4	Too et al. (2019)	A Comparative Study of Fine-Tuning Deep Learning Models for Plant Disease Identification	To compare different pre-trained CNN models for disease classification	ResNet50, VGG16, DenseNet, InceptionV4	Accuracy, Training Time, Validation Loss	54,000+ leaf images	DenseNet achieved highest accuracy of 99.75%

Table 3.1

Chapter 4

Existing Works & Implementation Works

4.1 Overview of Existing Work

Existing work related to plant leaf disease detection using Convolutional Neural Networks (CNNs) mainly focuses on detecting and classifying plant diseases using image processing and deep learning techniques. Earlier systems used traditional image processing methods such as image acquisition, preprocessing, segmentation, feature extraction, and classification for disease identification. However, with the advancement of Artificial Intelligence (AI) and Deep Learning, CNN-based approaches became more effective because they automatically extract important image features and provide higher prediction accuracy.

The proposed Plant Leaf Disease Detection System using CNN follows several important stages such as image acquisition, image preprocessing, data augmentation, CNN-based feature extraction, and disease classification. The overall workflow of the proposed system is explained below.

4.1.1 Image Acquisition

Image acquisition is the first and most important step in the plant disease detection process. In this project, plant leaf images were collected from the publicly available PlantVillage dataset. The dataset contains more than 54,000 RGB images belonging to 38 disease categories and 14 different plant species such as tomato, potato, grape, apple, pepper, and strawberry. The dataset includes both healthy and diseased plant leaves affected by bacterial, fungal, viral, and mold-based infections.

The collected images are stored digitally and used for further preprocessing and CNN model training. High-quality leaf images are essential for improving the overall performance and prediction accuracy of the deep learning model.

4.1.2 Image Preprocessing

Image preprocessing is performed to improve image quality and remove unnecessary distortions before training the CNN model. Preprocessing helps the model learn image patterns more effectively and improves classification accuracy. Several preprocessing techniques were applied in this project, including Image Resizing: All leaf images were resized into a fixed dimension suitable for CNN training.

Normalization: Pixel values were normalized to improve model convergence and training stability.

Data Augmentation: Techniques such as image rotation, flipping, zooming, shifting, and brightness adjustment were applied to increase dataset diversity and reduce overfitting.

Noise Reduction: Image enhancement methods were used to improve image clarity and feature visibility. These preprocessing operations improve the generalization capability of the CNN model and help classify leaf diseases under different environmental conditions.

4.1.3 Feature Extraction using CNN

Unlike traditional systems that use manual feature extraction techniques such as Gray-Level Co-occurrence Matrix (GLCM), the proposed system uses Convolutional Neural Networks (CNNs) for automatic feature extraction.The CNN model automatically learns important visual features from plant leaf images such as:

- Leaf spots
- Lesions
- Texture patterns
- Discoloration
- Mold growth
- Shape variations

The convolution layers extract low-level and high-level image features, while activation functions introduce non-linearity into the model. Max pooling layers reduce image dimensions and computational complexity while preserving important disease-related information.

Automatic feature extraction using CNN provides better performance compared to traditional handcrafted feature extraction techniques.

4.1.4. Classification

The final stage of the proposed system is disease classification. After extracting image features, the CNN model classifies plant leaves into different disease categories:

- Convolution layers
- ReLU activation layers
- Max pooling layers
- Flatten layer
- Dense layers
- Softmax output layer

The fully connected dense layers perform the final classification, while the softmax activation layer predicts probability scores for all disease classes. The disease category with the highest probability score is selected as the final prediction. The model was trained using the Adam optimizer and categorical cross-entropy loss function. Experimental results showed that the proposed CNN model achieved high prediction accuracy for classifying healthy and diseased plant leaves.

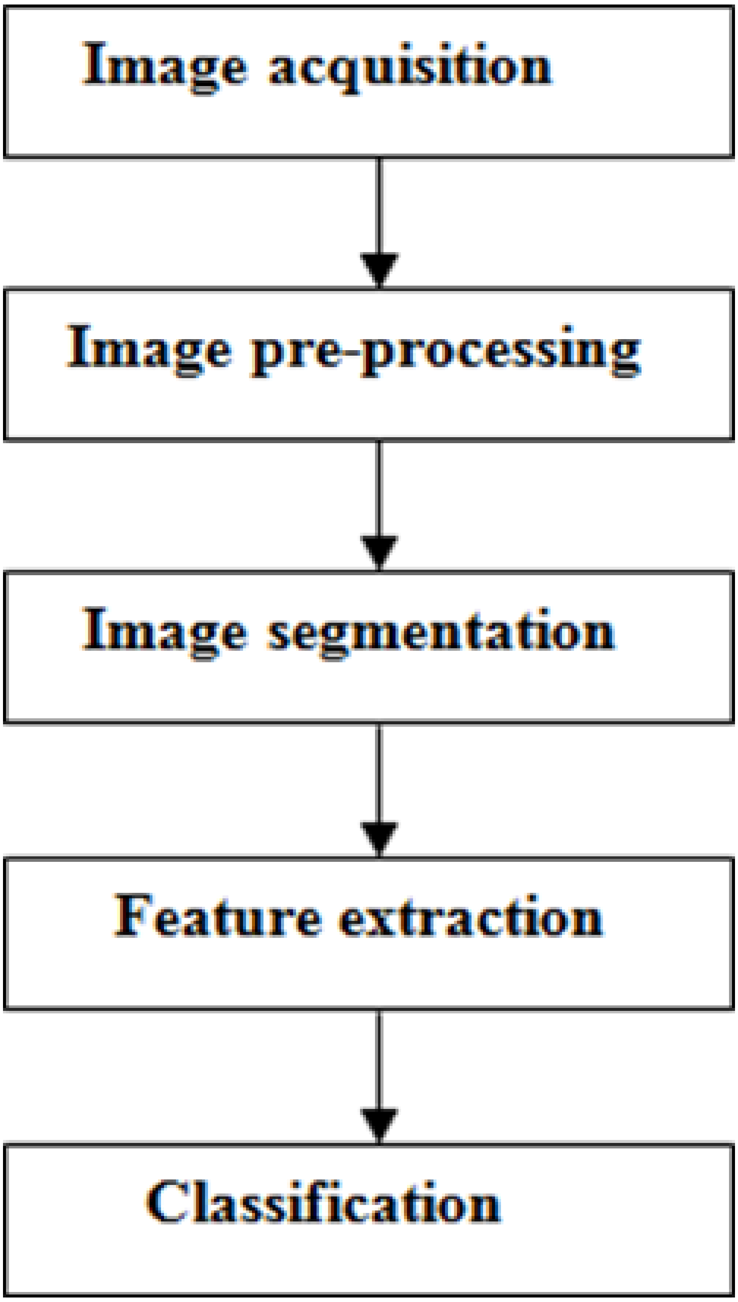

Fig:4.1 General block diagram of Feature Based Approach

4.2 Implementation Work

The proposed project titled "Plant Leaf Disease Detection System using CNN" was implemented using Deep Learning and image classification techniques for automatic detection and classification of plant leaf diseases. The system was developed using Python, TensorFlow, Keras, and Streamlit to provide an intelligent and user-friendly disease prediction platform for farmers and agricultural researchers.

In this project, the publicly available PlantVillage dataset was used for training and testing the CNN model. The dataset contains more than 54,000 RGB images across 38 disease categories and 14 different plant species. The dataset includes both healthy and diseased plant leaves from crops such as apple, tomato, potato, grape, corn, pepper, strawberry, peach, and cherry. The diseases included in the dataset are bacterial infections, fungal diseases, viral infections, mold diseases, rust, leaf spots, blight, and healthy leaf categories.

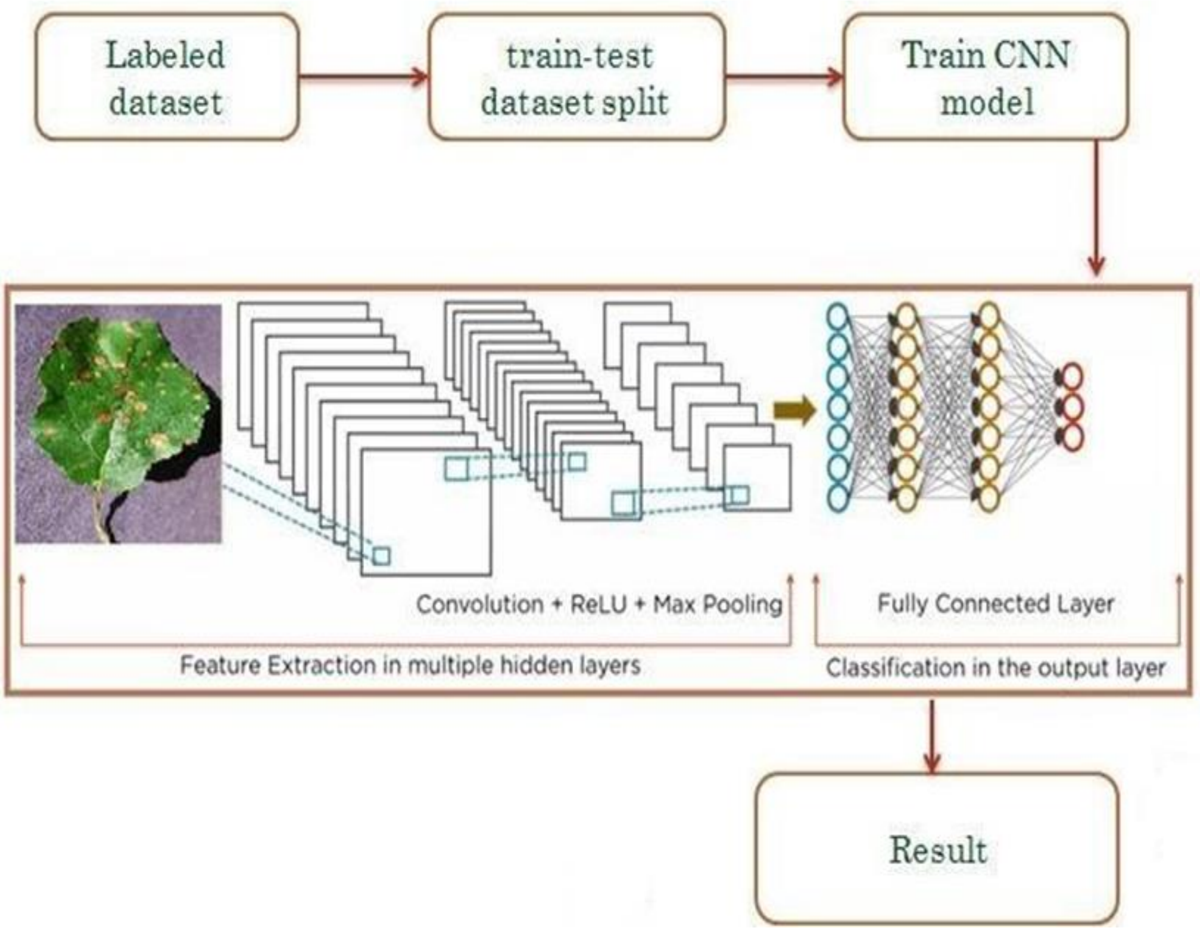

Fig: 4.2 Purposed workflow

The above figure represents the complete workflow and architecture of the Plant Leaf Disease Detection System using Convolutional Neural Network (CNN). The diagram illustrates how plant leaf images are processed through multiple stages including dataset preparation, feature extraction, CNN model training, and disease classification. The proposed architecture is designed to automatically identify and classify plant diseases from leaf images with high prediction accuracy. The entire workflow demonstrates the practical implementation of Deep Learning techniques in agriculture for intelligent crop disease monitoring.

The first stage shown in the figure is the Labeled Dataset. In this stage, plant leaf images are collected and organized into different disease categories. The dataset contains both healthy and diseased leaf images belonging to multiple plant species such as tomato, potato, grape, apple, corn, pepper, and strawberry. Each image in the dataset is assigned a label corresponding to a specific disease category. The labeled dataset is important because the CNN model learns disease patterns from these labeled examples during the training process. In this project, the PlantVillage dataset was used because it contains a large number of RGB leaf images across multiple disease classes. The dataset includes images affected by bacterial infections, fungal diseases, viral infections, rust, blight, and leaf spot diseases.

After collecting the labeled dataset, the next stage is the Train-Test Dataset Split. In this stage, the entire dataset is divided into two separate parts known as the training dataset and testing dataset. Generally, 80% of the images are used for training the CNN model, while the remaining 20% are used for testing and validation purposes. The training dataset helps the CNN model learn important disease features, while the testing dataset is used to evaluate the performance and prediction accuracy of the trained model. Splitting the dataset is an important step because it prevents overfitting and helps measure how well the model performs on unseen images.

Before feeding the images into the CNN model, image preprocessing operations are applied. Although preprocessing is not explicitly shown in the top workflow, it forms an essential part of the system. Preprocessing techniques such as image resizing, normalization, augmentation, and noise reduction are applied to improve image quality and enhance model performance. Image resizing converts all images into a fixed dimension suitable for CNN training. Normalization scales pixel values into a standard range, improving training stability and convergence speed. Data augmentation techniques such as flipping, rotation, zooming, and shifting increase dataset diversity and help the model generalize better under different environmental conditions.

The central part of the figure represents the Convolutional Neural Network (CNN) Architecture. CNN is one of the most powerful Deep Learning algorithms used for image classification and pattern recognition tasks. The CNN model automatically extracts important visual features from plant leaf images without requiring manual feature engineering. The architecture shown in the figure contains multiple hidden layers responsible for feature extraction and disease classification.

The first component inside the CNN architecture is the Convolution Layer. Convolution layers apply filters over the input leaf image to detect important visual patterns such as edges, spots, lesions, discoloration, and texture variations. These filters slide across the image and generate feature maps that capture disease-related information. The convolution operation enables the CNN model to learn low-level and high-level image features automatically. As the depth of the network increases, the model learns more complex disease patterns from plant leaves.

After the convolution operation, the output is passed through the ReLU Activation Function. ReLU, which stands for Rectified Linear Unit, introduces non-linearity into the network. Without activation functions, the CNN model would behave like a simple linear model and would not be able to learn complex disease patterns effectively. ReLU replaces negative values with zero while retaining positive values, improving training efficiency and reducing computational complexity. The activation layer also helps the model converge faster during training.

The next stage shown in the figure is the Max Pooling Layer. Pooling is used to reduce the size of feature maps generated by the convolution layers. The max pooling operation selects the maximum pixel value from small regions of the feature map and reduces dimensionality while preserving important image information. Pooling reduces computational cost, improves training speed, and makes the model less sensitive to small variations in the input image. In plant disease detection, pooling helps the CNN model focus on major disease patterns while ignoring unnecessary image details.

The figure indicates that multiple hidden layers consisting of convolution, ReLU, and max pooling operations are stacked together. These repeated layers help the CNN model learn increasingly complex disease features from leaf images. Early layers detect simple patterns such as edges and curves, while deeper layers identify disease-specific patterns such as leaf spots, mold growth, blight symptoms, and texture abnormalities.

The extracted features are then passed into the Fully Connected Layer. The fully connected layer performs the final classification task based on the features learned by the CNN model. In this stage, the multidimensional feature maps are converted into a one-dimensional vector through a flattening process. The fully connected neural network learns relationships between extracted features and disease categories. Dense layers inside the fully connected network process the features and generate prediction scores for different disease classes.

The final output layer uses the Softmax Activation Function for classification. Softmax computes probability scores for all disease categories and selects the disease class with the highest probability as the final prediction. The result generated by the model indicates whether the uploaded plant leaf is healthy or affected by a specific disease. The output is displayed to the user through the web application interface.

The last stage shown in the figure is the Result stage. After classification, the predicted disease category is displayed as the final result. The Streamlit-based web application allows users to upload plant leaf images and instantly receive disease prediction results. This real-time prediction system helps farmers detect plant diseases at an early stage and take preventive actions to reduce crop damage and improve productivity.

The architecture shown in the figure provides several advantages for plant disease detection. CNN-based systems eliminate the need for manual feature extraction and provide high classification accuracy. The model can automatically learn important disease patterns directly from images. The use of data augmentation and preprocessing techniques improves generalization and reduces overfitting. Real-time disease prediction through the Streamlit interface makes the system practical and user-friendly for agricultural applications.

Overall, the CNN-based architecture illustrated in the figure demonstrates the successful application of Deep Learning in smart agriculture. The proposed system provides an intelligent, automated, and efficient solution for plant disease detection. It supports precision farming, improves crop monitoring, reduces dependency on manual disease inspection, and contributes to sustainable agricultural development through AI-powered technologies.

Chapter 5

Dataset, Implementation and Result

5.1 Dataset Details

The success of any Deep Learning model mainly depends on the quality and size of the dataset used for training and testing. In the proposed project titled *"*Plant Leaf Disease Detection System using CNN*"*, the publicly available PlantVillage *Dataset* was used for developing and training the Convolutional Neural Network (CNN) model. The dataset contains a large collection of healthy and diseased plant leaf images captured under controlled environmental conditions.

The PlantVillage dataset is widely used in agricultural research and Deep Learning applications because it contains high-quality RGB images belonging to multiple crop species and disease categories. The dataset includes more than *54,000 plant leaf images* categorized into *38 disease classes* and 14 different plant species. These images represent various bacterial, fungal, viral, and mold-based diseases affecting agricultural crops.

The dataset includes plant species such as:

- Tomato
- Potato
- Apple
- Corn
- Grape
- Pepper
- Strawberry
- Peach
- Cherry
- Soybean
- Squash
- Raspberry

Each folder in the dataset contains images belonging to a specific disease category. Both healthy and diseased leaf images are included to help the CNN model distinguish between infected and non-infected leaves.

The disease categories included in the dataset are:

- Apple Scab
- Apple Black Rot
- Cedar Apple Rust
- Tomato Early Blight
- Tomato Late Blight
- Tomato Leaf Mold
- Tomato Mosaic Virus
- Tomato Yellow Leaf Curl Virus
- Potato Early Blight
- Potato Late Blight
- Corn Common Rust
- Corn Leaf Blight
- Grape Black Rot
- Leaf Spot Diseases
- Healthy Plant Leaves

and many other disease categories.

The dataset used in this project contains RGB images with different background conditions, textures, and disease symptoms. Disease symptoms include leaf spots, discoloration, lesions, fungal growth, rust formation, blight symptoms, and mosaic patterns.

5.1.1 Dataset Collection

The PlantVillage dataset was collected from publicly available agricultural image repositories. The dataset was downloaded and stored locally for preprocessing and model training purposes. The dataset was organized folder-wise according to disease classes.

The following steps were followed during dataset collection:

1. Downloading the PlantVillage dataset
2. Organizing images into class-wise folders
3. Removing corrupted images
4. Verifying image quality
5. Preparing the dataset for preprocessing

The dataset was then imported into the Python environment using image processing libraries.

5.1.2 Dataset Structure

The dataset structure plays an important role in Deep Learning model training. In this project, images were stored in separate folders based on disease labels.

Classes	no. of images	used images
Apple_scab	1000	1000
Apple_black_rot	1000	1000
Apple_cedar_apple_rust	1000	1000
Apple_healthy	1645	1000
Corn_gray_leaf_spot	1000	1000
Corn_common_rust	1192	1000
Corn_northern_leaf_blight	1000	1000
Corn_healthy	1162	1000
Grape_black_rot	1180	1000
Grape_black_measles	1383	1000
Grape_leaf_blight	1076	1000
Grape_healthy	1000	1000
Potato_early_blight	1000	1000
Potato_healthy	1000	1000
Potato_late_blight	1000	1000
Tomato_bacterial_spot	2127	1000
Tomato_early_blight	1591	1000
Tomato_healthy	1909	1000
Tomato_late_blight	1000	1000
Tomato_leaf_mold	1000	1000
Tomato_septoria_leaf_spot	1707	1000
Tomato_spider_mites_two-spotted_spider_mite	1676	1000
Tomato_target_spot	1404	1000
Tomato_mosaic_virus	1000	1000
Total images	30052	24000

Fig: 5.1 Dataset structure

5.1.3 Image Preprocessing

Before training the CNN model, several preprocessing operations were applied to improve image quality and model performance.

A. Image Resizing

All images were resized into a fixed dimension suitable for CNN model training. Image resizing ensures consistency throughout the dataset and reduces computational complexity.

B. Image Normalization

Normalization scales pixel values between 0 and 1, improving training stability and convergence speed.

C. Data Augmentation

Data augmentation techniques were applied to increase dataset diversity and prevent overfitting.

The augmentation techniques used include:

Rotation

Flipping

Zooming

Shifting

Brightness adjustment

D. Noise Reduction

Noise removal and enhancement techniques were applied to improve image clarity and disease visibility.

5.1.4 Train-Test Split

The dataset was divided into two parts:

Dataset Type	Percentage
Training Dataset	80%
Testing Dataset	20%

Table 5.1

The training dataset was used for CNN model learning, while the testing dataset was used to evaluate model performance on unseen data.

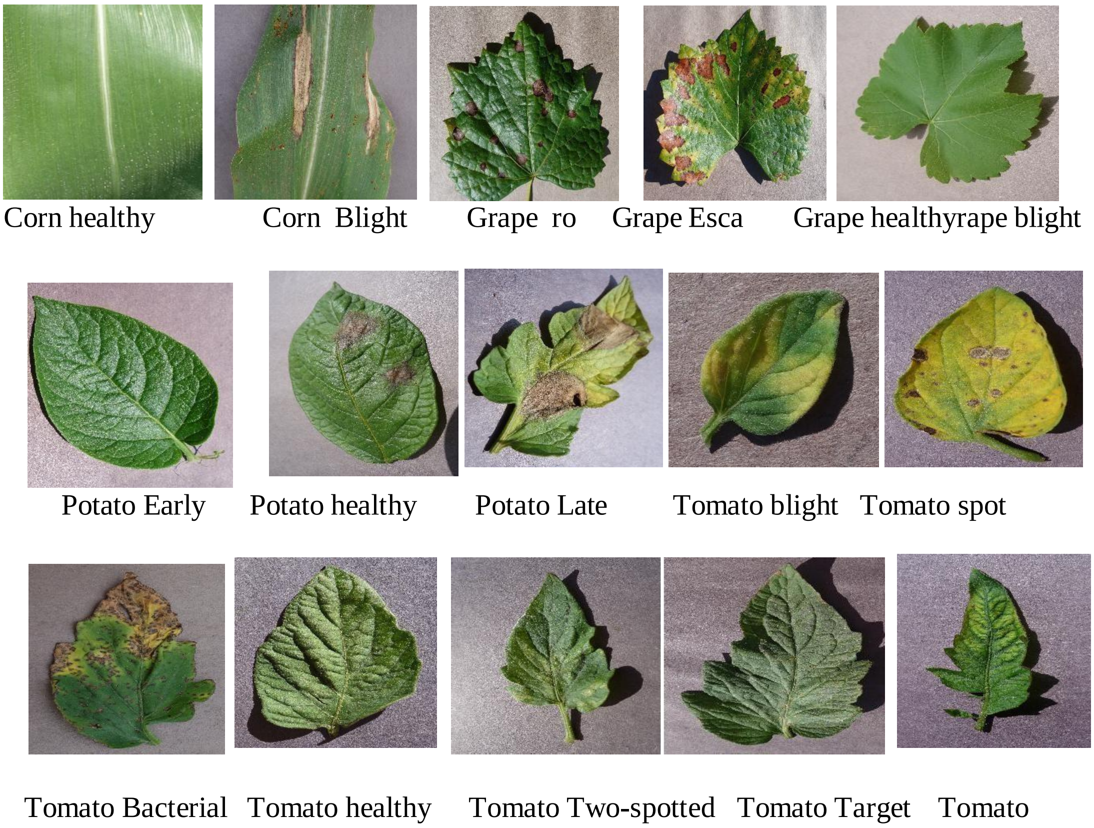

Fig: 5.3 Leaf images

5.2 Tools and Technology

The proposed project was implemented using various programming languages, Deep Learning frameworks, libraries, and software tools. These technologies were used for image processing, model training, evaluation, and deployment of the web application.

5.2.1 Python Programming Language

Python was used as the primary programming language for implementing the proposed Plant Leaf Disease Detection System using CNN because of its simplicity, flexibility, readability, and extensive support for Artificial Intelligence and Deep Learning applications. Python provides a rich ecosystem of libraries and frameworks that simplify image processing, model development, and data analysis tasks. In this project, Python was used for image preprocessing, dataset handling, CNN model training, evaluation, and deployment of the Streamlit web application. The language supports powerful libraries such as NumPy, TensorFlow, Keras, OpenCV, and Matplotlib, which are widely used in Machine Learning and Computer Vision projects. Python also supports GPU acceleration, enabling faster Deep Learning model training on large datasets like PlantVillage. Due to its easy syntax and strong community support, Python has become one of the most popular programming languages for developing AI-based agricultural applications and intelligent disease detection systems.

5.2.2 TensorFlow

TensorFlow is an open-source Deep Learning framework developed by Google that was used for building and training the CNN model in the proposed system. TensorFlow provides powerful functionalities for designing neural networks, performing tensor operations, optimizing model parameters, and handling large-scale computations efficiently. In this project, TensorFlow was used to train the Convolutional Neural Network using thousands of plant leaf images from the PlantVillage dataset. The framework supports convolution operations, activation functions, pooling layers, backpropagation algorithms, and optimization techniques required for Deep Learning applications. TensorFlow also provides GPU support, which significantly reduces model training time and improves computational performance. The framework helped the CNN model automatically learn disease-related features such as leaf spots, discoloration, lesions, rust formation, and fungal infections from plant leaf images. TensorFlow also supports model evaluation and visualization, allowing monitoring of training accuracy and loss during the learning process. Its scalability and high-performance computation capabilities made it highly suitable for implementing the plant leaf disease detection system.

5.2.3 Keras

Keras is a high-level Deep Learning API integrated with TensorFlow that was used for designing and implementing the CNN architecture in the proposed project. Keras provides a simple and user-friendly interface for creating neural network models efficiently with fewer lines of code. In this project, Keras was used for adding convolution layers, activation functions, max pooling layers, dropout regularization layers, flatten layers, and fully connected dense layers required for plant disease classification. The framework simplifies the process of model compilation, optimizer configuration, loss function selection, and model training. Keras also supports data augmentation and image preprocessing functionalities that improve model generalization and prevent overfitting during training. One of the major advantages of Keras is its modular architecture, which allows easy experimentation with different CNN configurations and hyperparameters. Keras helped reduce implementation complexity while maintaining high model performance and prediction accuracy. The integration of Keras with TensorFlow provided an efficient and powerful environment for developing the CNN-based Plant Leaf Disease Detection System.

5.2.4 NumPy

NumPy is a powerful Python library used for numerical computation and array processing in the proposed Plant Leaf Disease Detection System. In this project, NumPy was mainly used for handling multidimensional arrays and image data during preprocessing and CNN model training. Since digital images are represented as numerical pixel matrices, NumPy played an important role in converting plant leaf images into arrays suitable for Deep Learning operations. The library provides efficient mathematical functions and matrix operations that improve computational speed and memory management. NumPy was also used for normalization, reshaping image datasets, handling training and testing arrays, and performing mathematical computations required during model execution. Due to its high-performance array-processing capabilities, NumPy simplified image manipulation tasks and improved the efficiency of the CNN training process.

5.2.5 Matplotlib

Matplotlib is a widely used Python visualization library that was used in this project for plotting graphs, displaying images, and analyzing CNN model performance. Visualization is an important part of Deep Learning projects because it helps understand training progress and model behavior effectively. In the proposed system, Matplotlib was used to display sample plant leaf images, visualize disease categories, and generate accuracy and loss graphs during CNN model training. The library was also used to compare training accuracy and validation accuracy across multiple epochs. These visual graphs helped evaluate the learning performance of the CNN model and identify issues such as overfitting or underfitting. Matplotlib provides easy-to-use plotting functions that support data visualization in the form of line graphs, bar charts, image displays, and performance curves. The visualization capabilities of Matplotlib helped improve result analysis and made the experimental outcomes easier to interpret.

5.2.6 Google Colab

Google Colab was used as the development and execution environment for implementing and training the CNN model in the proposed project. Google Colab is a cloud-based Jupyter Notebook platform provided by Google that allows users to write and execute Python code directly through a web browser. One of the major advantages of Google Colab is its free access to GPU and TPU resources, which significantly improves Deep Learning model training speed. In this project, Google Colab was used for dataset uploading, preprocessing, CNN model development, training, evaluation, and visualization. The platform supports integration with popular libraries such as TensorFlow, Keras, NumPy, OpenCV, and Matplotlib, making it highly suitable for AI and Deep Learning applications. Google Colab also provides cloud storage integration through Google Drive, enabling easy access to large datasets like PlantVillage. The notebook environment helped simplify experimentation, debugging, and performance analysis during the implementation of the plant disease detection system.

5.2.7 Streamlit

Streamlit was used for developing the web-based graphical user interface (GUI) for real-time plant disease prediction. Streamlit is an open-source Python framework that allows developers to build interactive Machine Learning and Deep Learning web applications with minimal coding effort. In this project, Streamlit was integrated with the trained CNN model to provide a simple and user-friendly disease detection platform. The Streamlit application allows users to upload plant leaf images directly through the browser interface, after which the CNN model predicts the disease category and displays the result instantly. Streamlit supports rapid deployment of AI applications and provides interactive components such as buttons, file uploaders, text displays, and image rendering features. The framework simplified the deployment process and made the proposed system accessible to farmers, researchers, and agricultural users without requiring technical knowledge. The real-time prediction capability provided by Streamlit improved the practical usability of the Plant Leaf Disease Detection System and demonstrated the successful deployment of Deep Learning models in smart agriculture applications.

5.3 Result

```
, Epoch 1/10
1087/1087 ━━━━━━━━━━━━━━━━━━━━ 77s 63ms/step - accuracy:
0.5208 - loss: 1.7288 - val_accuracy: 0.7669 - val_loss: 0.7807
Epoch 2/10
1087/1087 ━━━━━━━━━━━━━━━━━━━━ 67s 61ms/step - accuracy:
0.7285 - loss: 0.8851 - val_accuracy: 0.8284 - val_loss: 0.5578
Epoch 3/10
1087/1087 ━━━━━━━━━━━━━━━━━━━━ 65s 59ms/step - accuracy:
0.8053 - loss: 0.6124 - val_accuracy: 0.8726 - val_loss: 0.4121
Epoch 4/10
1087/1087 ━━━━━━━━━━━━━━━━━━━━ 83s 61ms/step - accuracy:
0.8470 - loss: 0.4773 - val_accuracy: 0.8701 - val_loss: 0.4147
Epoch 5/10
1087/1087 ━━━━━━━━━━━━━━━━━━━━ 64s 59ms/step - accuracy:
0.8775 - loss: 0.3770 - val_accuracy: 0.9025 - val_loss: 0.3235
Epoch 6/10
1087/1087 ━━━━━━━━━━━━━━━━━━━━ 67s 62ms/step - accuracy:
0.8939 - loss: 0.3166 - val_accuracy: 0.9045 - val_loss: 0.3160
Epoch 7/10
1087/1087 ━━━━━━━━━━━━━━━━━━━━ 80s 60ms/step - accuracy:
0.9081 - loss: 0.2752 - val_accuracy: 0.9092 - val_loss: 0.3088
Epoch 8/10
1087/1087 ━━━━━━━━━━━━━━━━━━━━ 67s 62ms/step - accuracy:
0.9176 - loss: 0.2507 - val_accuracy: 0.9058 - val_loss: 0.3577
Epoch 9/10
1087/1087 ━━━━━━━━━━━━━━━━━━━━ 67s 62ms/step - accuracy:
0.9284 - loss: 0.2169 - val_accuracy: 0.9070 - val_loss: 0.3460
Epoch 10/10
1087/1087 ━━━━━━━━━━━━━━━━━━━━ 65s 60ms/step - accuracy:
0.9328 - loss: 0.1983 - val_accuracy: 0.9072 - val_loss: 0.3729
<keras.src.callbacks.history.History at 0x7bd4a8d7ce00>
```

Fig: 5.4 Training Model accuracy

```
340/340 ━━━━━━━━━━━━━━━━━━━━ 11s 33ms/step - accuracy: 0.9081
- loss: 0.3433
[0.3433212339878082, 0.908111572265625]
```

Fig: 5.5 Testing model accuracy

Model: "sequential"

Layer (type)	Output Shape	Param #
rescaling (Rescaling)	(None, 224, 224, 3)	0
conv2d (Conv2D)	(None, 222, 222, 32)	896
max_pooling2d (MaxPooling2D)	(None, 111, 111, 32)	0
conv2d_1 (Conv2D)	(None, 109, 109, 64)	18,496
max_pooling2d_1 (MaxPooling2D)	(None, 54, 54, 64)	0
conv2d_2 (Conv2D)	(None, 52, 52, 128)	73,856
max_pooling2d_2 (MaxPooling2D)	(None, 26, 26, 128)	0
flatten (Flatten)	(None, 86528)	0
dense (Dense)	(None, 128)	11,075,712
dropout (Dropout)	(None, 128)	0
dense_1 (Dense)	(None, 38)	4,902

Total params: 33,521,588 (127.87 MB)
Trainable params: 11,173,862 (42.62 MB)
Non-trainable params: 0 (0.00 B)
Optimizer params: 22,347,726 (85.25 MB)

Fig: 5.6 CNN Model Summary

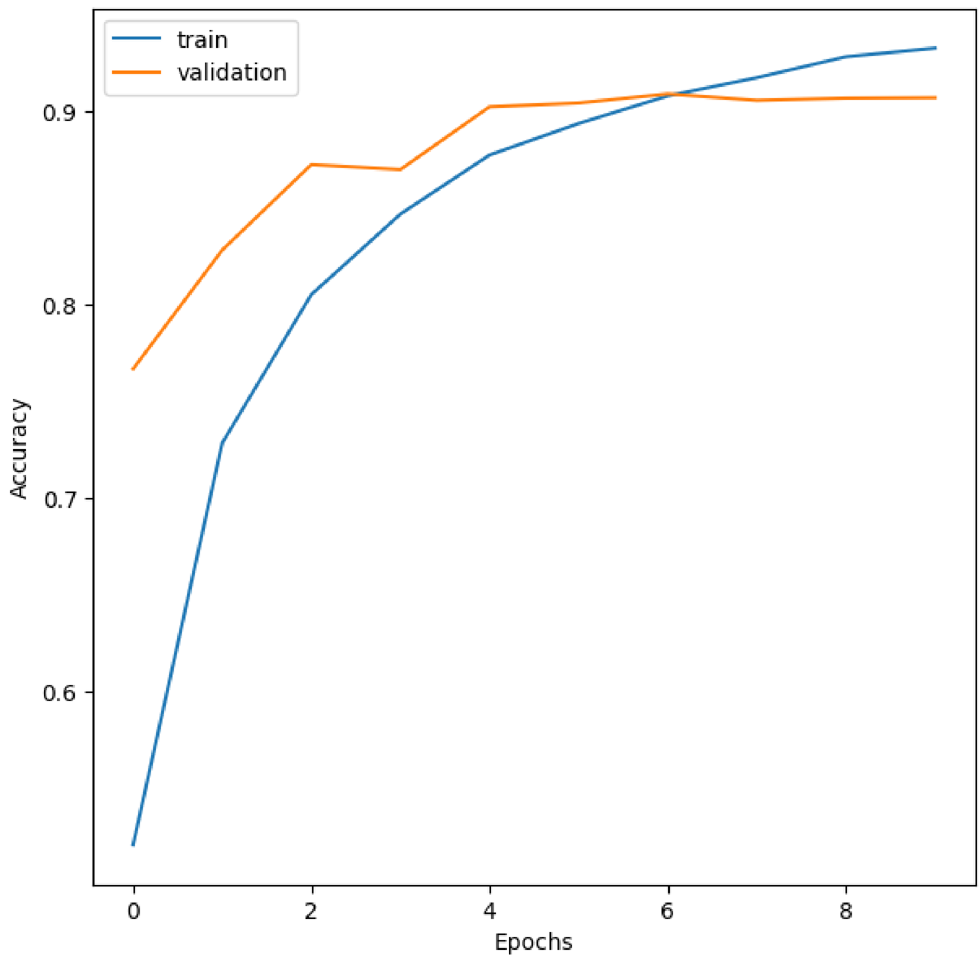

Fig: 5.7 Training accuracy and validation accuracy

At the beginning, the training accuracy starts around 52%, while the validation accuracy is already higher at approximately 76%. As the epochs increase, the training accuracy improves significantly and reaches around 94% by the final epoch. Similarly, the validation accuracy also increases and stabilizes around 90–91%.

The close gap between training and validation accuracy indicates that the CNN model is performing well and generalizing effectively on unseen data. Around epoch 6 onward, the validation accuracy becomes stable, while the training accuracy continues to increase slightly. This suggests that the model has nearly converged and there is only a small amount of overfitting.

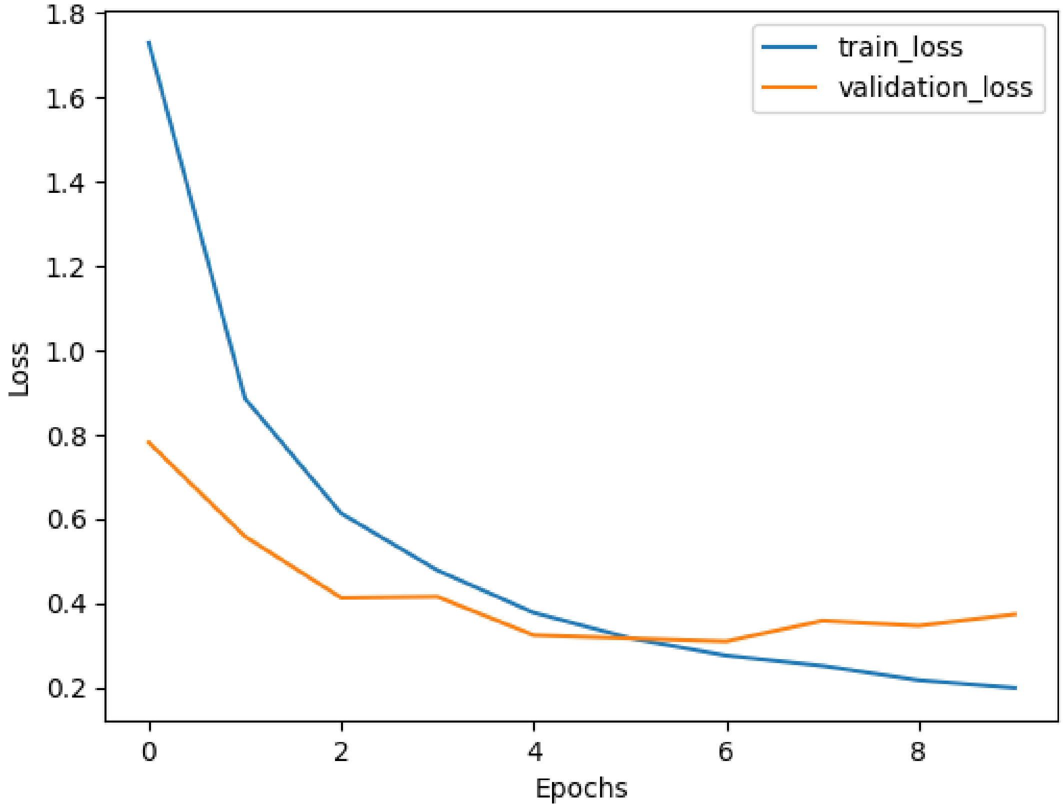

Fig: 5.8 Training loss and validation loss

At the initial stage, the training loss is relatively high at around 1.7, but it decreases rapidly as the epochs progress, finally reaching approximately 0.20. Similarly, the validation loss starts near 0.78 and gradually decreases to around 0.31–0.37.

The decreasing trend of both training and validation loss shows that the CNN model is successfully minimizing errors during the learning process. Around epochs 5–6, the validation loss reaches its minimum value and then shows slight fluctuations, while the training loss continues to decrease. This indicates a small sign of overfitting after later epochs, but the difference between the two curves remains relatively small.

Fig: 5.9Confusion matrix

The figure represents the Confusion Matrix of the CNN-based Plant Disease Detection model. A confusion matrix is used to evaluate the performance of a classification model by comparing the actual class labels with the predicted class labels.

In this graph:

The rows represent the true labels (actual plant diseases).

The columns represent the predicted labels generated by the CNN model.

The values inside the matrix indicate the number of predictions made for each class.

The darker blue color intensity represents higher prediction counts.

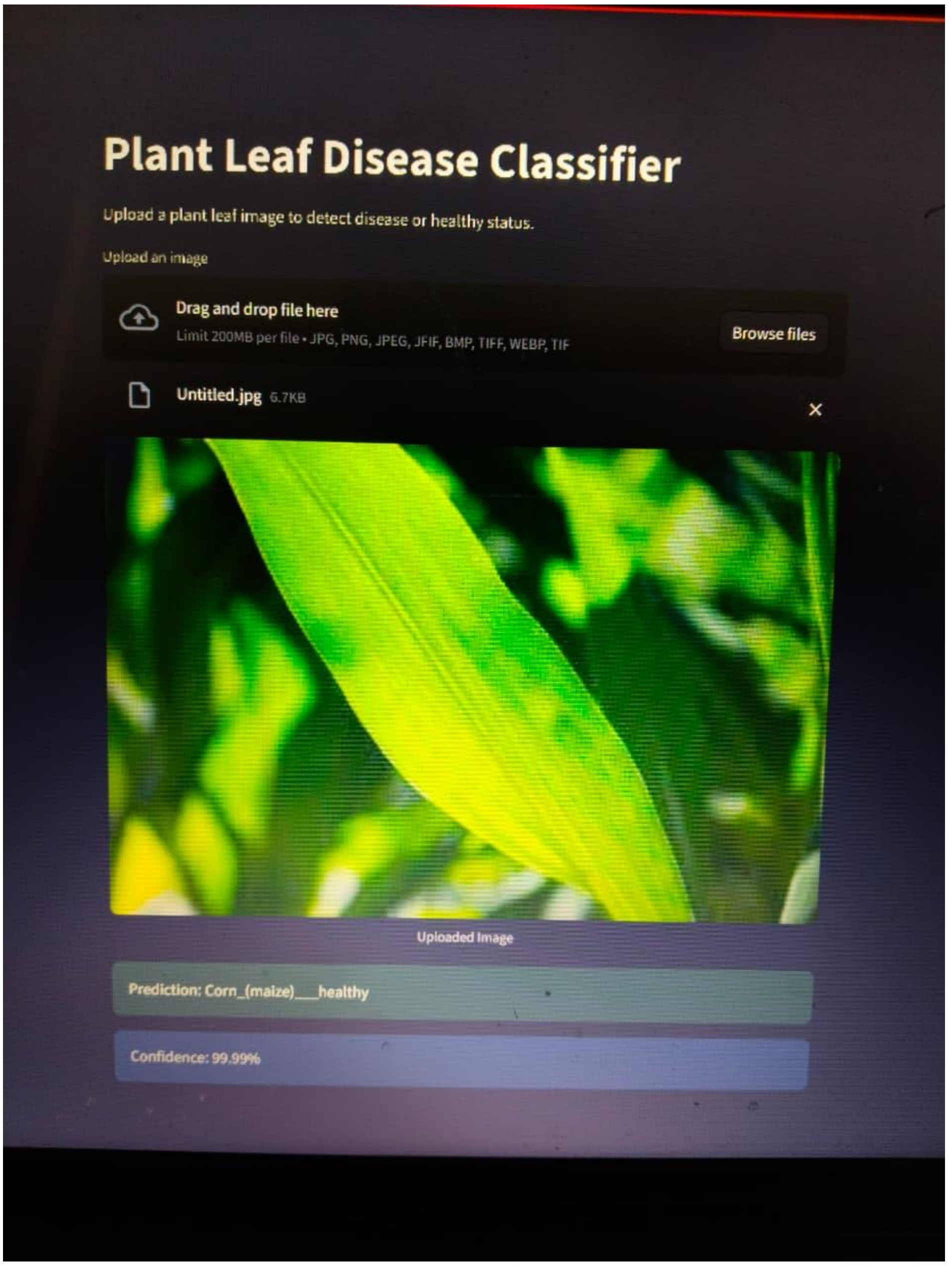

Fig: 5.10 User interface

The figure shows the user interface of the Plant Leaf Disease Classifier system developed using Streamlit and a CNN deep learning model. The application allows users to upload plant leaf images and automatically predict whether the leaf is healthy or affected by a disease.

In the displayed example, a maize (corn) leaf image has been uploaded into the system through the drag-and-drop file upload feature. After processing the image, the CNN model predicts the class as "Corn_(maize)__healthy" with a confidence score of 99.99%.

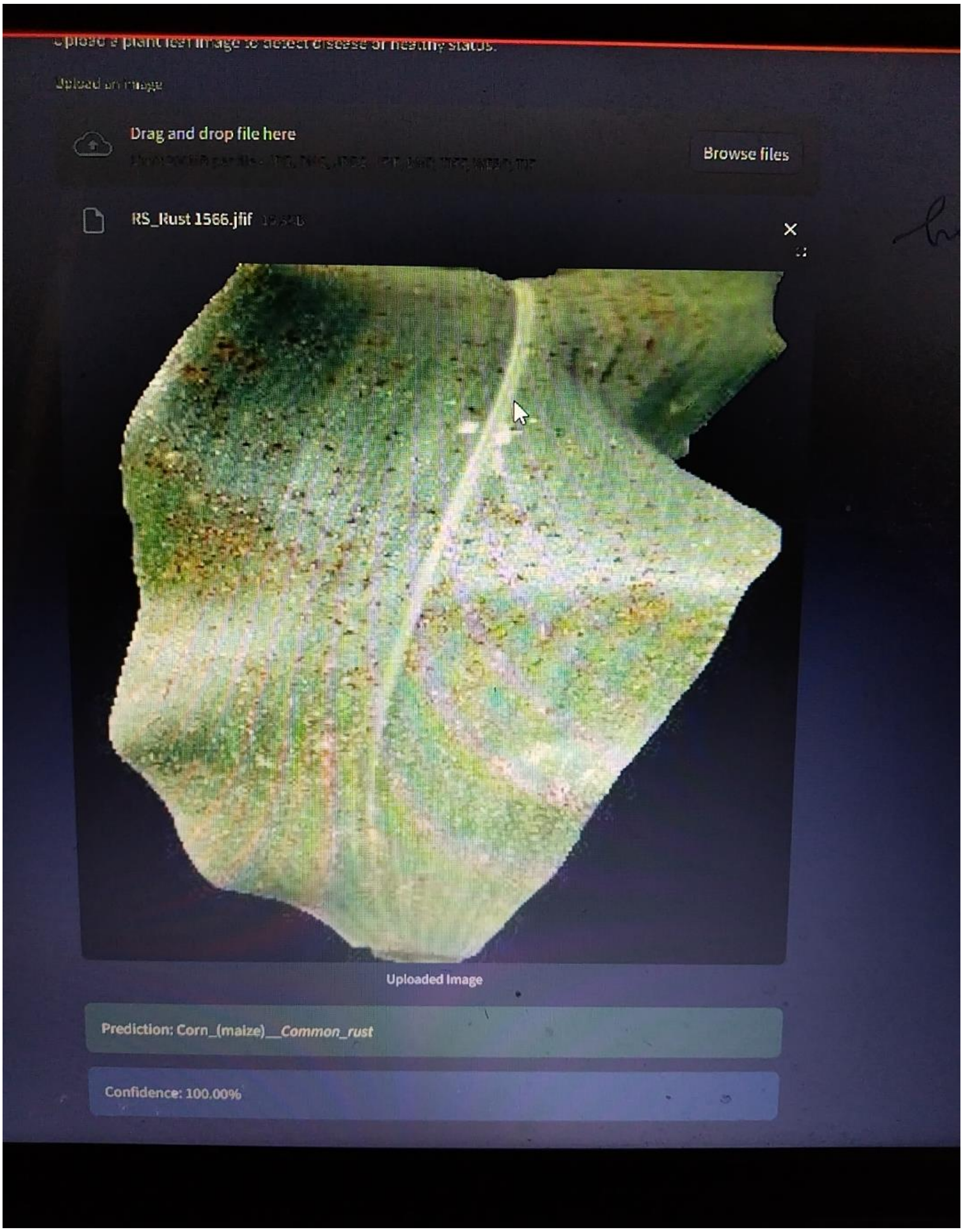

Fig: 5.11 User interface

The figure demonstrates the working output of the Plant Leaf Disease Classifier developed using a CNN deep learning model and Streamlit framework. In this example, the user uploads an image of a maize leaf affected by disease into the application interface.

After analyzing the uploaded image, the CNN model predicts the disease as "Corn_(maize)__Common_rust" with a confidence score of 100.00%

Chapter 6

Conclusion

The Plant Disease Detection System using Convolutional Neural Network (CNN) was successfully developed to identify and classify plant leaf diseases using deep learning techniques. The main objective of this project was to provide an intelligent, accurate, and automated solution for detecting plant diseases at an early stage through image processing and CNN-based classification.

In this project, a CNN model was trained on a large dataset of plant leaf images containing both healthy and diseased leaves from multiple crop categories. Various preprocessing techniques such as image resizing, normalization, and data augmentation were applied to improve the robustness and performance of the model. The trained model achieved high training and validation accuracy with low loss values, demonstrating effective learning and good generalization capability.

The confusion matrix and prediction results confirmed that the model can accurately classify different plant diseases with minimal misclassification. A user-friendly web application was also developed using Streamlit, allowing users to upload plant leaf images and receive real-time disease predictions along with confidence scores. The system successfully identified diseases such as Common Rust, Early Blight, Leaf Spot, and healthy leaf conditions with high prediction confidence.

This project demonstrates the practical application of Artificial Intelligence and Deep Learning in smart agriculture. The developed system can assist farmers, agricultural researchers, and crop management organizations in early disease diagnosis, reducing crop loss and improving productivity. In the future, the system can be enhanced by integrating mobile applications, real-time camera detection, IoT sensors, and advanced deep learning architectures for even higher accuracy and scalability.

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